

Worked Answer Key

Page 1. 1. Certain: a die always gives one of 1–6. 2. Blue; $9/10$ is greater than $1/10$.

Page 2. 1. {heads, tails}. 2. {1, 2, 3, 4}.

Page 3. 1. One favorable outcome out of six: $1/6$. 2. Even outcomes are 2, 4, 6: $3/6 = 1/2$.

Page 4. 1. Total $4 + 3 + 1 = 8$; $P(\text{yellow}) = 3/8$. 2. Not purple: $7/8$.

Page 5. 1. 0; blue is impossible because none is in the bag. 2. $12/12 = 1$; certain.

Page 6. 1. Not orange: $5/8$. 2. $1 - 1/5 = 4/5$.

Page 7. 1. A: $2/8 = 1/4$; B: $4/12 = 1/3$; Bag B is greater. 2. $3/10 = 0.30$, $1/4 = 0.25$; $3/10$ is greater.

Page 8. 1. Yes. Heads and tails each have probability $1/2$, so each player has an equal chance. 2. No. Red is $3/4$ and blue is $1/4$, so chances are unequal.

Page 9. 1. $P(6) = 1/6$; $60 \div 6 = 10$, so about 10 sixes. 2. Green is $5/10 = 1/2$; $40 \div 2 = 20$.

Page 10. 1. $17/30$. 2. $9/25$.

Page 11. 1. $18/120 = 3/20$; $600 \times 3/20 = 90$. 2. $12/20 = 3/5$; $100 \times 3/5 = 60$.

Page 12. 1. Total $6 + 5 + 3 + 2 = 16$; $P(\text{green}) = 3/16$. 2. $P(\text{red or blue}) = (6 + 5)/16 = 11/16$.

Page 13. 1. {HH, HT, TH, TT}; one favorable outcome, so $1/4$. 2. HT and TH are favorable: $2/4 = 1/2$.

Page 14. 1. Color pairs are RR, RB, BR, BB; $P(RR) = (2/3)(2/3) = 4/9$. 2. $P(RB \text{ or } BR) = (2/3)(1/3) + (1/3)(2/3) = 4/9$.

Page 15. 1. No. Probability must be from 0 to 1; $7/4 > 1$. 2. The claim is false. $1/2$ describes long-run expectation; a small number of trials can differ.

Page 16. 1. Total 12; $P(\text{apple}) = 5/12$ (5 favorable apples over 12 fruits). 2. 0 means no favorable outcomes, such as rolling a 7 on a six-sided die. 1 means every outcome is favorable, such as rolling a number from 1–6 on that die.

Page 17. 1. Books or games: $3 + 2 = 5$ favorable spaces out of 10, so $1/2$. 2. Other fruit: $24 - 15 = 9$; $9/24 = 3/8$.

Page 18. 1. Use 3 equal sections labeled for the three players; each has $1/3$, so it is fair. 2. Use 4 equal sections with 1 blue and 3 other colors; blue is $1/4$.

Page 19. 1. Total 12; not green is $3 + 4 = 7$, so $7/12$. 2. Odd outcomes (1, 3, 5) have probability $3/6 = 1/2$; $90 \times 1/2 = 45$, assuming a fair die and similar independent rolls.

Page 20. 1. Greater than 4: {5, 6}, so $2/6 = 1/3$. Red tile: $3/10$. Since $1/3 > 3/10$, rolling greater than 4 is more likely. 2. Theoretical probability = favorable equally likely outcomes $\div$ total equally likely outcomes. Less than 3 means 1 or 2: $2/6 = 1/3$.